B. K. Birla College of Arts, Science and Commerce, Kalyan (W)

TYBSc Physics, Semester VI Special Theory of Relativity

Assignment 1

Unit 1: Why Relativity? Postulates, Simultaneity, Time Dilation and Length Contraction

Name: _____ Roll No.: _____ Date: _____

Total marks: 60 **Each question carries 5 marks**

Instructions: Attempt all questions. Section A: theory (write to the point, with neat diagrams where useful). Section B: numerical problems and derivations (state formulae used, show all steps and give units). Sign convention for the interval: $s^2 = \Delta x^2 + \Delta y^2 + \Delta z^2 - c^2\Delta t^2$; four-vectors $a^\mu = (a^0, \mathbf{a})$ with $a \cdot b = -a^0 b^0 + \mathbf{a} \cdot \mathbf{b}$.

Section A: Theory (6 × 5 = 30 marks)

Q1. What is an inertial frame of reference? Write the Galilean transformation equations for two frames in the standard configuration and state the principle of Newtonian (Galilean) relativity. Which physical quantities are invariant under the Galilean transformation and which are not? **[5]**

Q2. Explain why the laws of electromagnetism are incompatible with Newtonian relativity. Discuss the three alternatives that were open to physicists around 1900 and state which one was finally accepted. **[5]**

Q3. What was the ether hypothesis? List the properties the ether was required to have. Explain how Bradley's stellar aberration ruled out the hypothesis that the Earth drags the ether along with it. **[5]**

Q4. Describe the Michelson–Morley experiment: its purpose, the apparatus (with a labelled diagram) and the procedure. What result was expected, what was observed, and what is the significance of the result? **[5]**

Q5. Discuss the contribution of H. A. Lorentz to the origin of relativity: the contraction hypothesis, "local time" and the Lorentz transformation. How did Einstein's interpretation of the same equations differ from Lorentz's? **[5]**

Q6. State Einstein's two postulates of the special theory of relativity. Using Einstein's train thought experiment, explain how they lead to the relativity of simultaneity. **[5]**

Section B: Numericals and Derivations (6 × 5 = 30 marks)

Q7. Show that Newton's second law is invariant under the Galilean transformation, but that the wave equation $\dfrac{\partial^2\psi}{\partial x^2} - \dfrac{1}{c^2}\dfrac{\partial^2\psi}{\partial t^2} = 0$ is not. **[5]**

Q8. Derive an expression for the fringe shift expected in the Michelson–Morley experiment when the interferometer is rotated through $90°$. Calculate its value for an effective arm length of $11\,\text{m}$, $\lambda = 590\,\text{nm}$ and an orbital speed of the Earth of $3 \times 10^4\,\text{m/s}$. Show that the Lorentz–FitzGerald contraction makes the expected shift zero. **[5]**

Q9. Derive the time-dilation formula using a light clock. Charged pions have a proper mean life of $2.6 \times 10^{-8}\,\text{s}$. For a pion beam moving at $0.9c$, find (a) the mean life measured in the laboratory and (b) the mean distance travelled before decay. **[5]**

Q10. Muons (proper mean life $2.2\,\mu$s) are produced at a height of 9.0 km and travel vertically down at $0.995c$. Find (a) their mean life in the Earth frame, (b) the mean distance they travel, (c) the thickness of the 9.0 km of atmosphere in the muons' frame, and (d) the fraction of muons that reach sea level. **[5]**

Q11. Derive the length-contraction formula. A spaceship of proper length 100 m flies past the Earth at $0.8c$. (a) What length does an Earth observer measure? (b) How long does the ship take to pass a fixed point on the Earth, according to the Earth observer? **[5]**

Q12. (a) A metre stick at rest in S' makes an angle of 60° with the x'-axis. S' moves at $0.6c$ along x. Find the length of the stick and the angle it makes with the x-axis in S. (b) Two lamps 600 m apart along the x-axis flash simultaneously in S. Find the time interval between the flashes in a frame moving at $0.8c$ along x, and say which flashes first. **[5]**

Useful constants: $c = 3.0 \times 10^8$ m/s; $e = 1.6 \times 10^{-19}$ C; $m_e c^2 = 0.511$ MeV; $m_p c^2 = 938.3$ MeV; $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m; $\mu_0 = 4\pi \times 10^{-7}$ H/m; $1/(4\pi\varepsilon_0) = 9.0 \times 10^9$ N m^2/C^2; $G = 6.67 \times 10^{-11}$ N m^2/kg^2; $M_\odot = 1.99 \times 10^{30}$ kg; 1 ly $= 9.46 \times 10^{15}$ m; 1 pc $= 3.086 \times 10^{16}$ m $= 3.26$ ly; 1 yr $= 3.156 \times 10^7$ s; $b_{\rm Wien} = 2.898 \times 10^{-3}$ m K; $k_B = 8.62 \times 10^{-5}$ eV/K; $a = 7.566 \times 10^{-16}$ J m^{-3} K^{-4}.